

MAGNETIC EFFECT OF ELECTRIC CURRENT

- **Introduction**

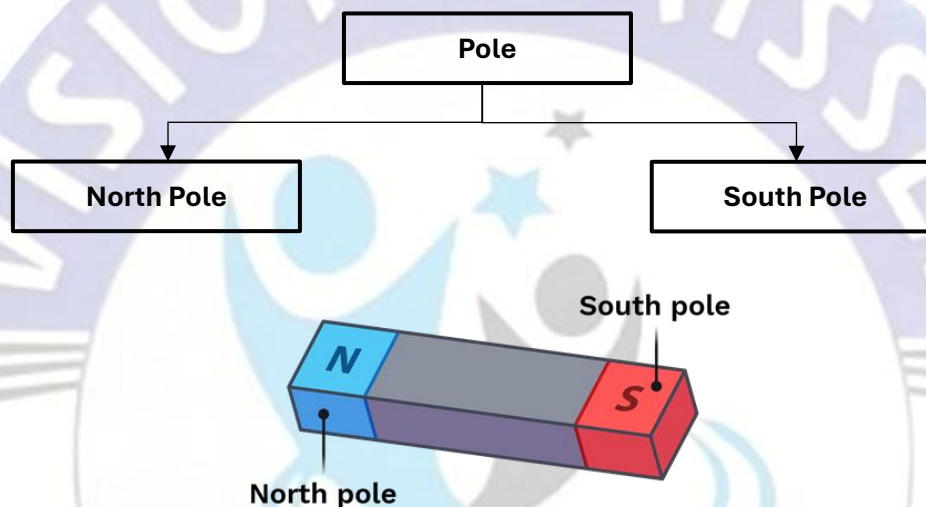
- Magnetic effect of electric current is defined as "when an electric current flowing in a conductor produces Magnetic Field around it.
- Electric current can produce magnetism.
- Electric current can produce heating effect / electric field / magnetic field.

- **Magnet**

- A magnet is that substance which attracts pieces of iron, nickel and cobalt items.

- **Poles of magnet**

- That end point of magnet where the attraction force of magnet is very high is called poles of magnet.

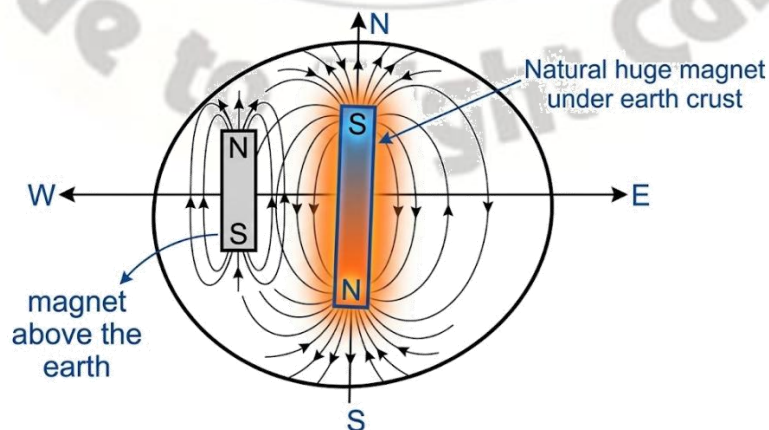


1. **North Pole of the magnet**

- That end of a freely hanged magnet which ends point forward the north direction is called the north Pole of the magnet.

2. **South Pole of the magnet**

- That end of a freely hanged magnet which ends point forward the south direction is called the South Pole of the magnet



- **Magnetic axis**

- The line joining the north & south pole of the magnetic is called magnetic axis.

Note:

1. Like poles repel to each other.

{S – S / N – N}

→ Repel

2. Unlike poles attract each other.

{S – N / N – S}

→ Attract

- **Magnetic Substances**

- Those substances which attract towards the magnet called magnetic substance.

For example:-

- ✓ Iron, Cobalt, Nickel & Its alloy.

- **Non – magnetic substance**

- Those substance which do not attract toward the magnet called non – magnetic substance.

For example:-

- ✓ Glass, Paper, Plastic, Brass etc.

- **Bar Magnet**

- Those magnet which found in cuboidal shape are called bar magnet.

- **Magnetic Compass**

- A magnetic compass is a simple device in which a magnet like as the size of a needle which is framed in non – magnetic frame the needle can move freely in a horizontal plane. The north pole of the needle points towards the geographical north of the earth.



Types of Magnets

1. Permanent Magnet.
2. Hard Magnetic Material.

1. **Permanent Magnet**

- Those magnets which made of certain alloys of iron, nickel, cobalt, retain their magnetism for a long time are called permanent magnet.

2. **A hard Magnetic material**

- Those magnets which made of alloy of aluminum nickel, cobalt and certain type of steel which retains magnetism for a short time is called A hard magnetic material.

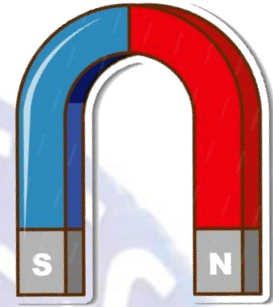
- **Magnet of different shapes**



Bar Magnet



Circular Magnet



Horse shoe Magnet

- **Magnetic field**

- That field around a magnet in which magnet attracts or repels any magnetic substance / article called magnetic field of that magnet. It is denoted by B (Baber).
- The S.I unit of 'B' is tesla and it is denoted by T.

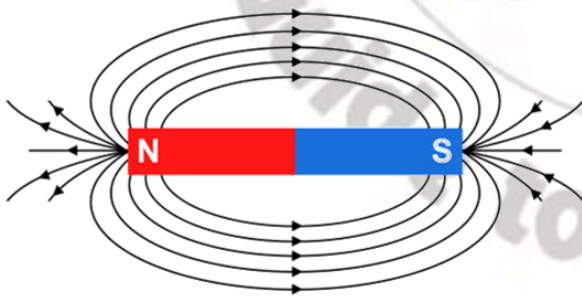
- **Magnetic Force**

- That force by which any magnets attracts any magnetic substance towards it, called magnetic force.
- Or,
The Force applied by magnetic field on magnetic substance is called magnetic force.

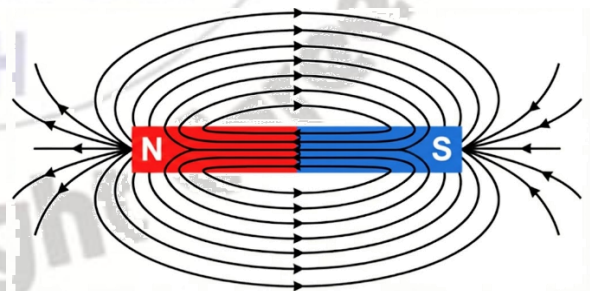
- **Magnetic Field lines**

- The magnetic field which is expressed those lines are called magnetic field lines.

- **An experiment in which magnetic field lines are expressed**



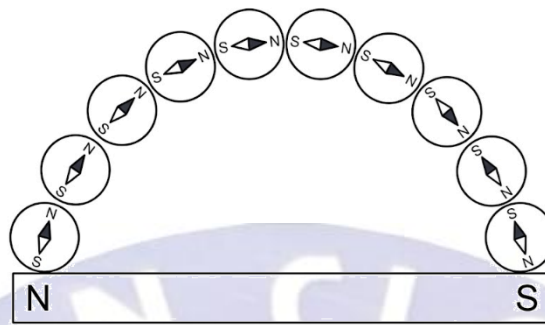
External magnetic field lines



Internal magnetic field lines

- Magnetic field lines emit from the north pole of magnet rod and passes through curve path ended on south pole.

- **Magnetic field lines are expressed by compass needle method.**



- **Properties of magnetic field line**

- All field lines are closed curves they are came out of the magnet from the side of the north pole and go into it on the side of the south pole. They continue inside the magnet too (Parallel to the magnetic axis).
- The field lines are close together near the poles and spread out away from them. The field is stronger where the field lines are more closely spaced. So, the field is stronger near the poles then at other point.
- The field lines never intersect each other. If two field lines where to intersect them a compass needle placed at the point of intersection would points in two direction – which is not possible.
- The line such that the tangents at any points on it gives the direction of the magnetic field.

- **Magnetic Field created by electric current**

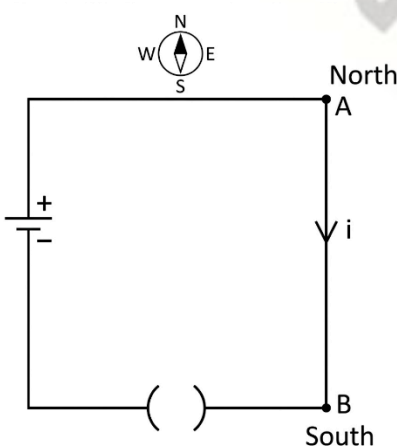
Hans Christian Orested

In 1820,

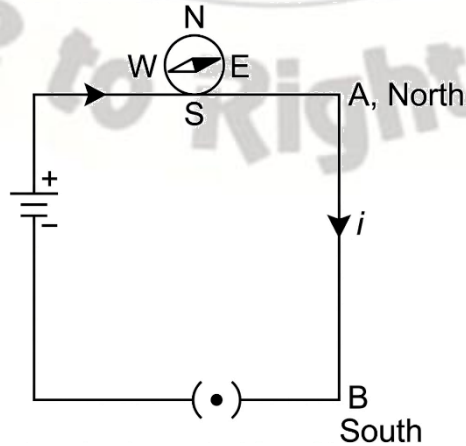
Orested was first observed that when an electric current flowing in a conductor produces a magnetic field around it.

The magnetic effect of electric current can be very easily demonstrate by bringing a magnetic compass near a current carrying wire. The compass needle gets deflected showing that a magnetic field is produced near the wire.

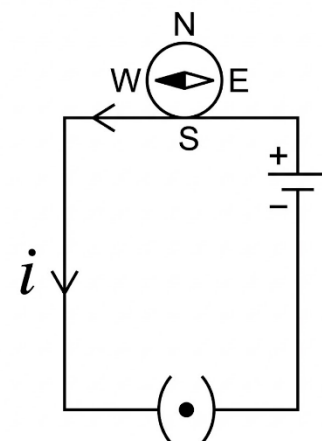
- **The experiment is shown in the figure**



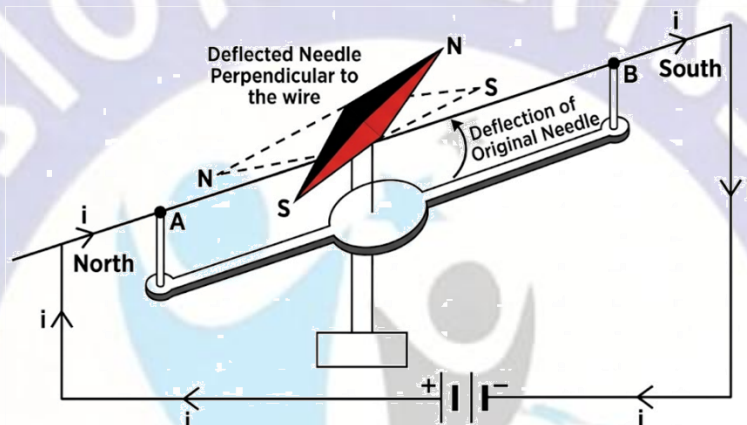
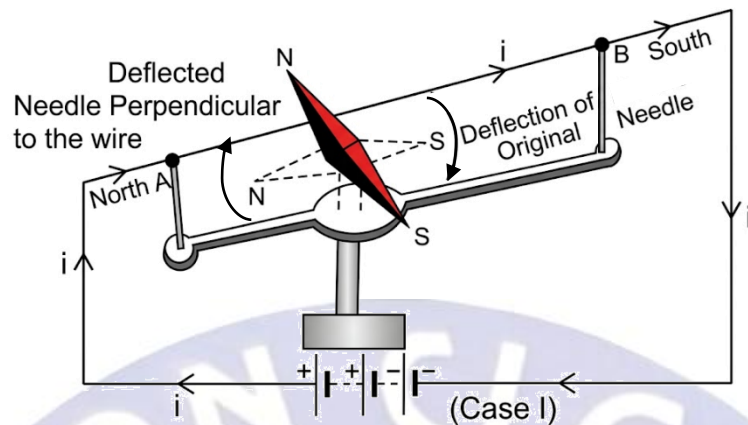
Case 1



Case 2



Case 3



- The shape of magnetic field lines produced by current carrying conductor depends on its shape.

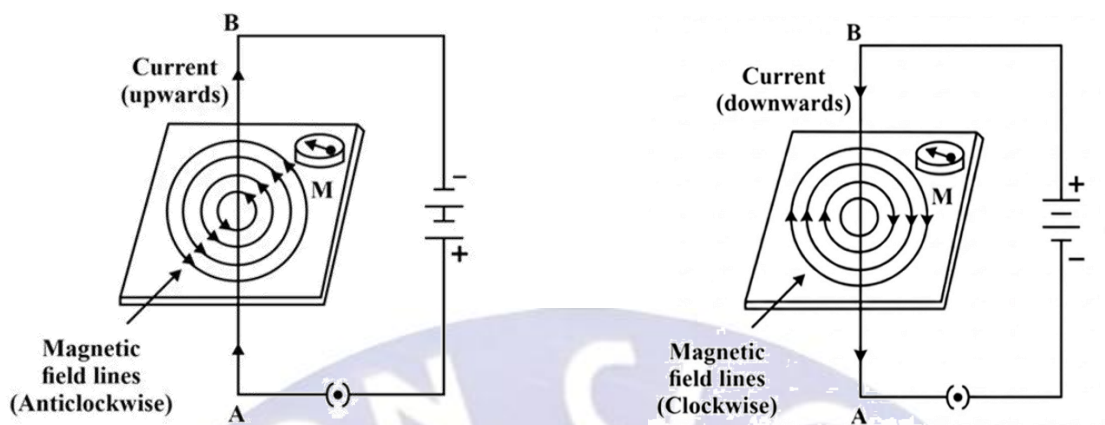
- So, different shape of conductor having different shape of magnetic field line.
- There are three types of magnetic field lines when the shape of conductor or wire
 - i) A straight wire carrying current.
 - ii) A circular loop carrying current.
 - iii) A solenoid & long coil wire carrying current.

- Magnetic field lines due to current carrying straight wire

- In this experiment, First we fix a long stiff wire AR upright in a piece of cardboard kept horizontally connected the wire a battery through a switch as show in below figure. Now place a compass on the cardboard. Start the current by closing the switch the compass needle will get deflected, and its direction will show the direction of the magnetic field at that point.

Mark the position of north pole of the needle on the cardboard. Shift the compass ahead so that the south pole of needle lies at the point marked for the previous position of the north pole repeat the procedure till you reach the starting point. Now we joint a marked point. Thus, the circular lines gives magnetic field lines.

- If the electric current is strong the lines will be nearly circular.
- If the electric current is weak it will produce a weak magnetic field or away circular.



- The magnetic field lines are found in concentric circle.
- If the direction of current is reversed the magnetic field will be still circular, but the direction of field lines which is reversed.

Note:

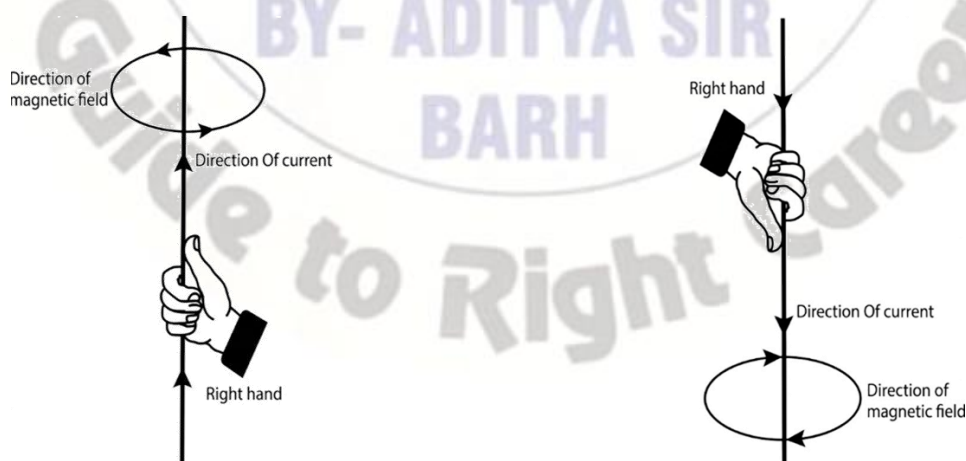
1. The magnetic field lines $\propto$ current passing in the wire.
2. The magnetic field lines $\propto \frac{1}{\text{distance of that point from the wire}}$.

• **Maxwell's Right – hand thumb Rule**

- The direction of the magnetic field due to a current in a straight conductor is given by Maxwell's right – hand thumb rule.

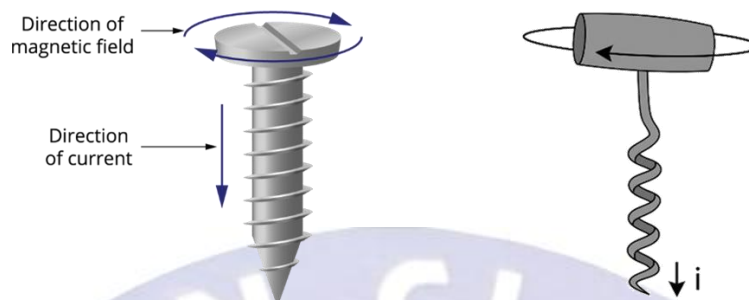
The rule states as:

If a straight current carrying wire is imagined to be held in the right hand with the thumb stretched along the direction of the current, the magnetic field produced by the current is in the direction in which the rest four fingers are curled.



- This rule is also known as Maxwell's Cork Screw rule.

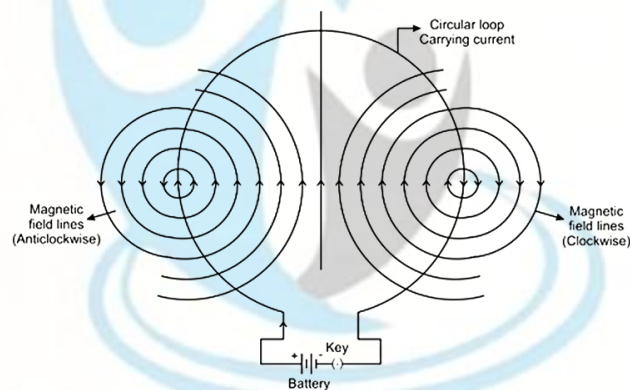
Suppose, the tip of rotating corkscrew or a crew advantages in the direction of the current.



- **Magnetic field line shape due to a circular loop**

- Let a copper wire that form a circle. It is fixed vertically in a piece of cardboard, with half the loop above the cardboard and below it. Gain the free ends of the wire to a battery through a switch and pass a current through it.

We observed that the line will nearly concentric circles centered on the point through which the wire passes. The line near the center at a loop is perpendicular to the plane of the loop with the help of iron fillings.

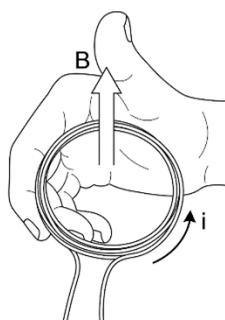


The magnetic field lines above the cardboard

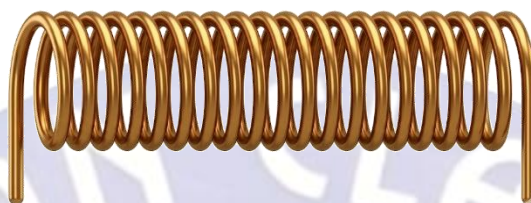
- **Right hand palm rule for loop**

- A coil is made up by winding several turns of insulated wire in a circular form. When a current passes through a coil, each turn produces its own magnetic field. These fields together give rise to a net magnetic field that is stronger than that produced by a circular loop of single turn.
- If the coil has n turn, the magnetic field due to the coil is n time stronger than due to single turn.

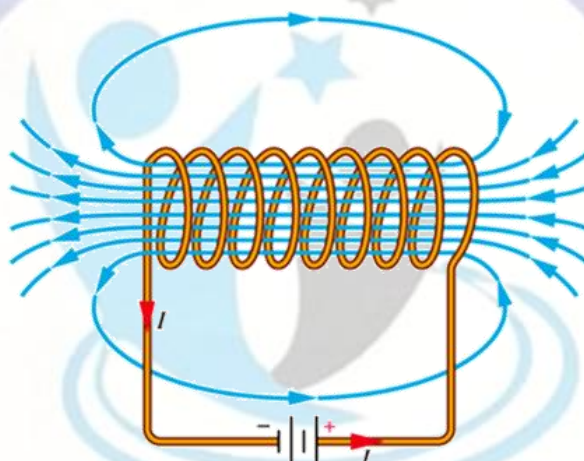
1. Strength of magnetic field of circular coil $\propto$ No. of turn of insulated wire
2. Strength of magnetic field of circular coil $\propto$ current passing through the conductor
3. Strength of magnetic field of circular coil $= \frac{1}{\text{Radius of the coil}}$



- If the fingers of the right hand are curled along the direction at the center in a loop, the stretched thumb gives the direction of the magnetic field.
- **Magnetic field due to solenoid (long coiling)**
- When a long-insulated wire is tightly wound in the shape of a spring, with closely spaced turns that lie side by side is called solenoid.



- The wire is winding over a non-conducting cylinder tube with differ from a circular coil when enters electric current in a solenoid, then solenoid produces a magnetic field.



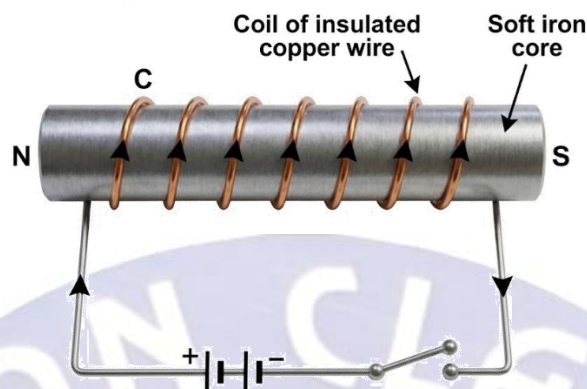
- The magnetic field lines of a solenoid is similar to that of a bar magnet.
- The magnetic field lines inside the solenoid is parallel to the axis of solenoid.
- The magnetic field increases near the end.
- The magnetic field of a current carrying wire is not uniform. It changes with distance from the center.
- If we holding the solenoid in hand with four finger curved along the direction of the current, then the direction of the thumb gives the direction of magnetic field.
- **The strength of magnetic field produced by a current carrying solenoids on.**
 1. The strength of magnetic field $\propto$ the no. of turns in the solenoid
 2. The strength of magnetic field $\propto$ the strength of current in the solenoid
 3. The strength of magnetic field $\propto$ the nature of core material

Note:

- Soft iron rod as a core in a solenoid produced the strongest magnetism.

- **Electromagnet**
- An electromagnet is a temporary magnet consisting of a long coil of insulated copper wire wrapped around a soft iron core.
- It is magnetised only when electric current is passed through the coil.

- The core of electromagnet must be of soft iron because soft iron easily magnetises and demagnetises.
- Steel is not used at the place of soft iron because steel does not lose all its magnetism when current is off and it becomes a permanent magnet.



Note:

- Soft iron is the pure form of iron which is used in making nails, bolts etc.

▪ **Factors affecting strength of electromagnetism**

1. The strength of electromagnetism $\propto$ the no. of turns in the coil.
2. The strength of electromagnetism $\propto$ the current flowing in the coil.
3. The strength of electromagnetism $\propto \frac{1}{\text{The length of gap between it.}}$

▪ **The difference between permanent magnet & electromagnet**

Bar magnet / Permanent Magnet	Electromagnet
The bar magnet is a permanent magnet.	An electromagnet is a temporary magnet. Its magnetism is only for the current passing through it. So, the magnetism of an electromagnet can be switched on or switched off as desired.
A permanent magnet produces a comparatively weak force of attraction.	An electromagnet can produce very strong magnetic force.
The strength of a permanent magnet cannot be changed.	The strength of an electromagnet can be changed by changing the current passing through it.
The (north – south) polarity of a permanent magnet is fixed and cannot be changed.	The polarity of an electromagnet can be changed by changing the direction of current in its coil.

• **The Factors which reduce the magnetism of magnet**

- There are three main causes which can affect the magnetism of a permanent magnet.

1. **Heat**

- Heating of magnets above the Curie temperature causes the magnetic domains to be disrupted (चुम्बकिय भुसमप्ती में दरार आना) permanently. Mild heating causes reduction in the magnetism. However, when it cools the reduced magnetism is restored.

Domain $\rightarrow$ Area of work / Activity

2. Mechanical shock

- A magnet that is subjected to shock such as being hit by a hammer or dropped from a height can lose its magnetism. However, modern magnets made from materials such as samarium, cobalt and neodymium can withstand shock.

3. Hitting by a hammer

- When we hit the magnet with a hammer all the little magnet fairies. In the metal get hurt and runaway to hide. They cut all of their invisible strings.

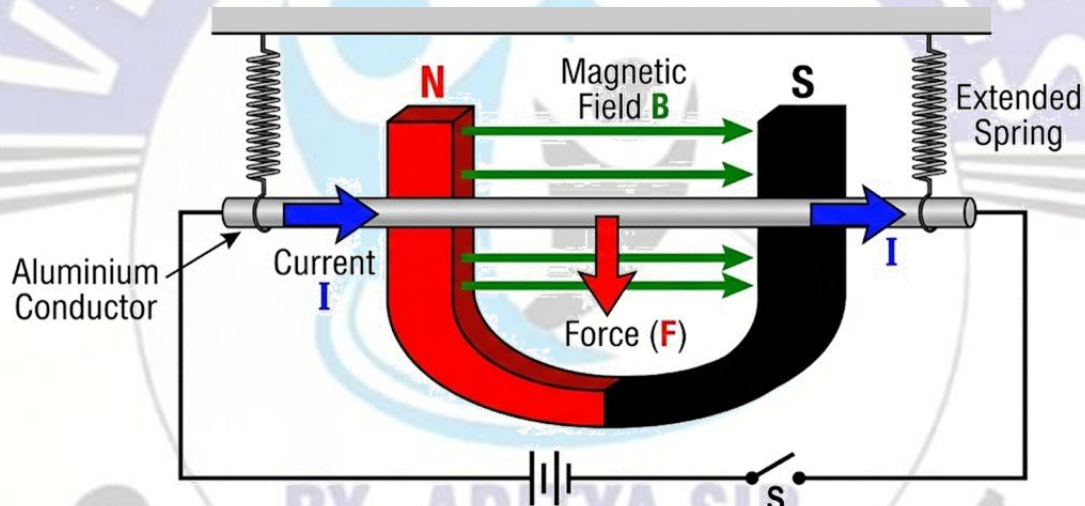
Note:

- When we supply an A.C current in a permanent magnet then magnet reduce its magnetism.

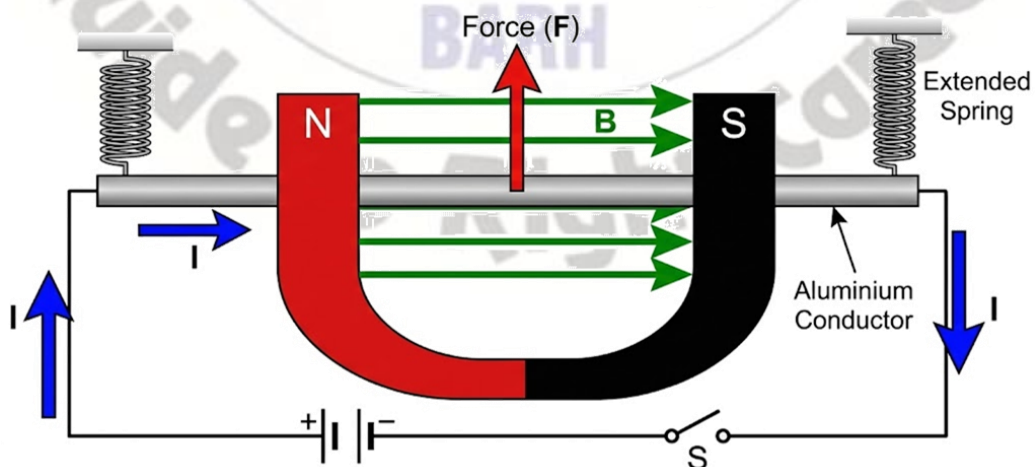
- **A magnetic field exerts a force on a current carrying conductor**

- In this experiment we take a aluminum conductor, a switch and a horse-shoe magnet and all are arranged in the below picture.

Case 1:



Case 2:

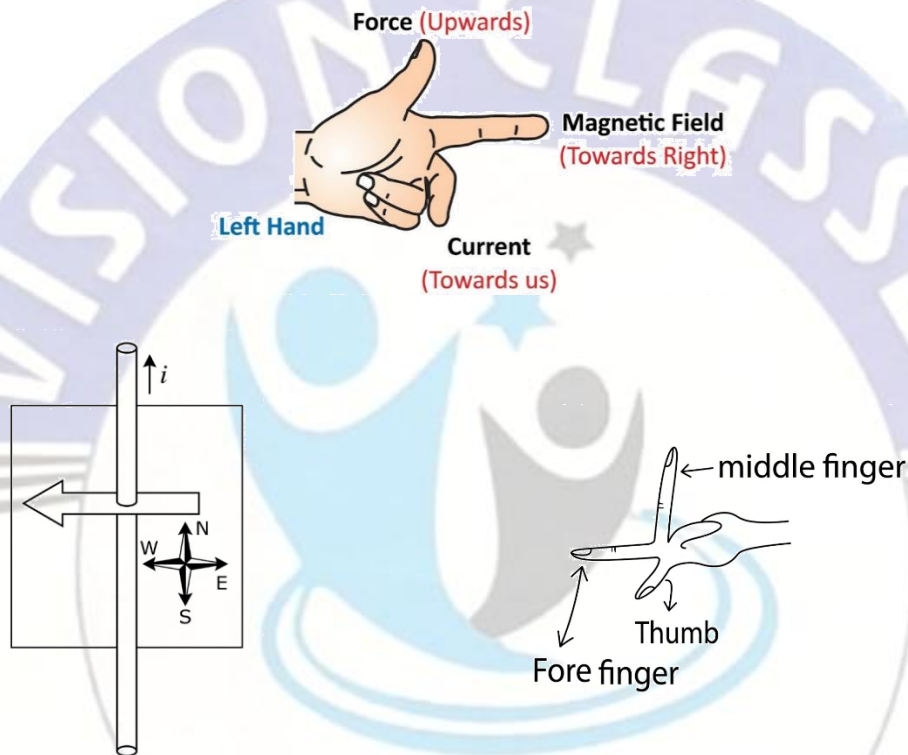


▪ In this experiment we see that

1. A current carrying conductor experience a force in a magnetic field.
2. The direction of the force depends on the direction of the magnetic field and perpendicular to it.
3. The direction of the force depends on the direction of the current and is perpendicular to it.

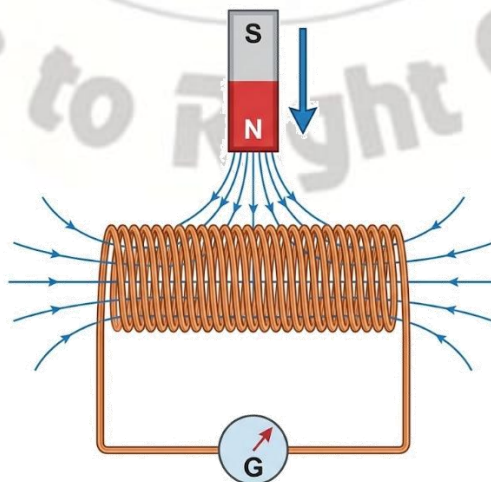
• Fleming's Left-hand Rule

- If the forefingers second finger and thumb of the left hand are stretched at right angles to each other with the forefinger in the direction of the field and the second finger in the direction of the current then the thumb indicates the direction of the force.



• Electromagnetic induction

- That phenomenon in which an electric current is induced in a circuit because of changing magnetic field is called electromagnetic inductions.



▪ **We conclude the following experiment:-**

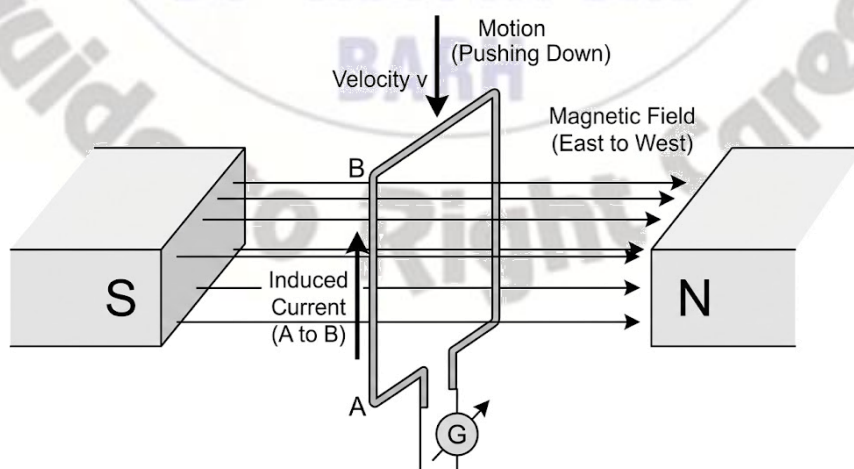
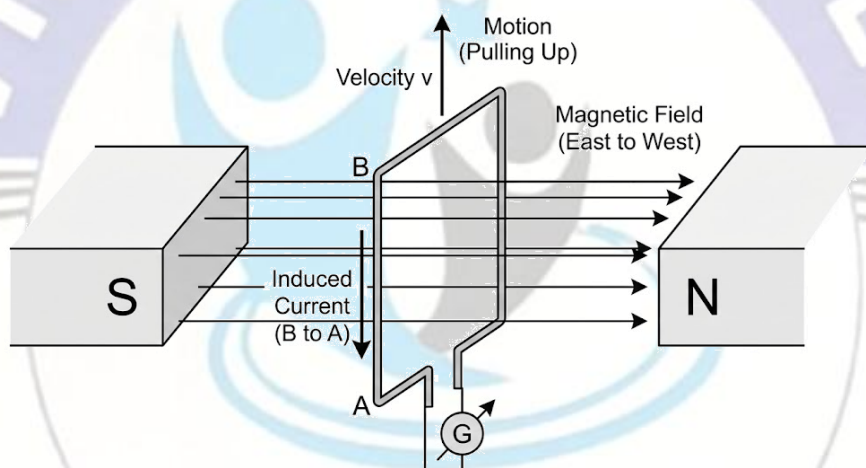
- A current flows in the coil when there is relative motion between the magnet and the coil.
- The changing magnetic field causes an induced current in the coil.
- The direction of the current depends on the direction of the magnetic field.
- The direction of the current depends on whether the magnetic field around the coils increasing or decreasing.

Note:

- Not only we can use an electric current to produce a magnetic field, we can also use a magnetic field to produce an electric current.

• **Fleming's Right-hand Rule**

- If the fore finger, middle finger and thumb at the right hand are stretched at right angles to each other, with the fore finger in the direction of the field and the thumb in the direction of the motion of the wire, then the second finger gives the direction of induced current.

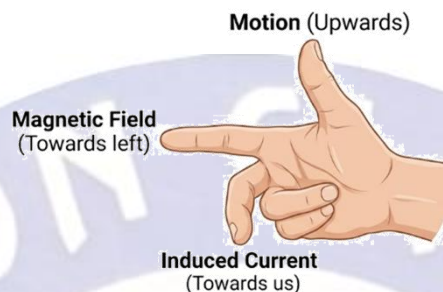


- wire AB is connected with a galvanometer, the magnetic field is from east to west and the wire AB is perpendicular to the field. i.e, it lies in the south – north direction.

- a) Suppose the loop is pulled vertically a current will be included in the loop in the direction from B to A.
- b) Suppose the loop is pushed down vertically, the current will be induced in the loop current from A to B.

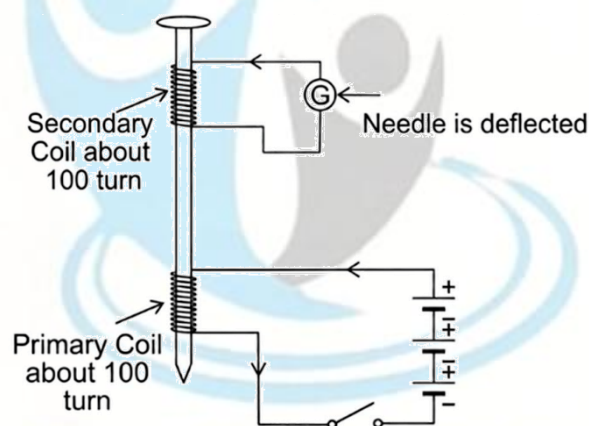
Note:

- **Fleming's Right-hand Rule** gives the direction of the induced current. When a section of a loop passes through a field perpendicular it.



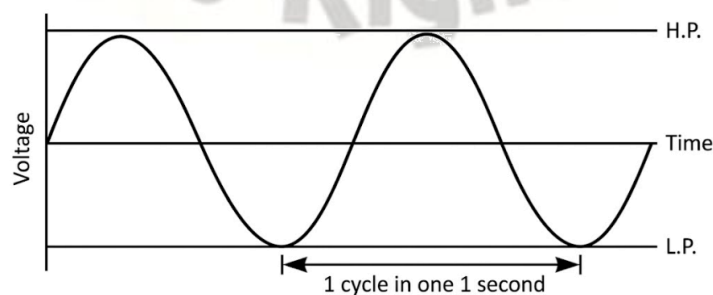
Fleming's Right-hand rule

- **Inducing a current without a magnet**



- **Alternating current (A.C)**

- That current which doesn't flow in the same direction. It changes rapidly every second through an electric device is called an Alternating current.
- The frequency of our household supply is 50Hz. i.e, 50 cycles / second. It means it changes direction 100 times in 1 second.
- Symbol of an A.C current is

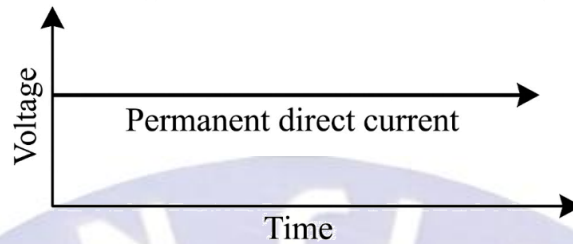


For example:-

- ✓ That current which flows in our houses, official schools etc.

- **Direct Current (D.C)**

- That current which flows always flows in the same direction through an electrical devices is called a direct current.



For example:-

- ✓ The electric current flows by the battery.

Case 1: When current passes through a straight wire.

$$B = \frac{\mu_0 I}{2\pi R}$$

where,

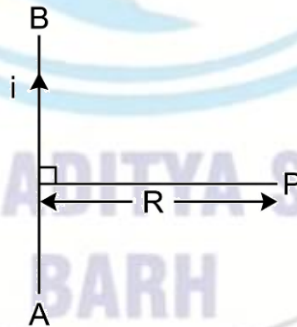
B = magnetic field

μ_0 = Permeability of vacuum (constant) [Value of $\mu_0 = 4 \times 10^{-7} \text{ Fm}^{-1}$]

I = amount of electric current

R = $\perp$ distance from conductor

→ S.I unit of μ_0 is F/m or Fm^{-1}



Case 2: When current passes through in circular wire / coil.

$$B = \frac{\mu_0 \cdot N \cdot I}{2R}$$

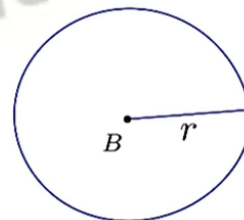
where,

μ_0 = Permeability of vacuum

N = no. of turns

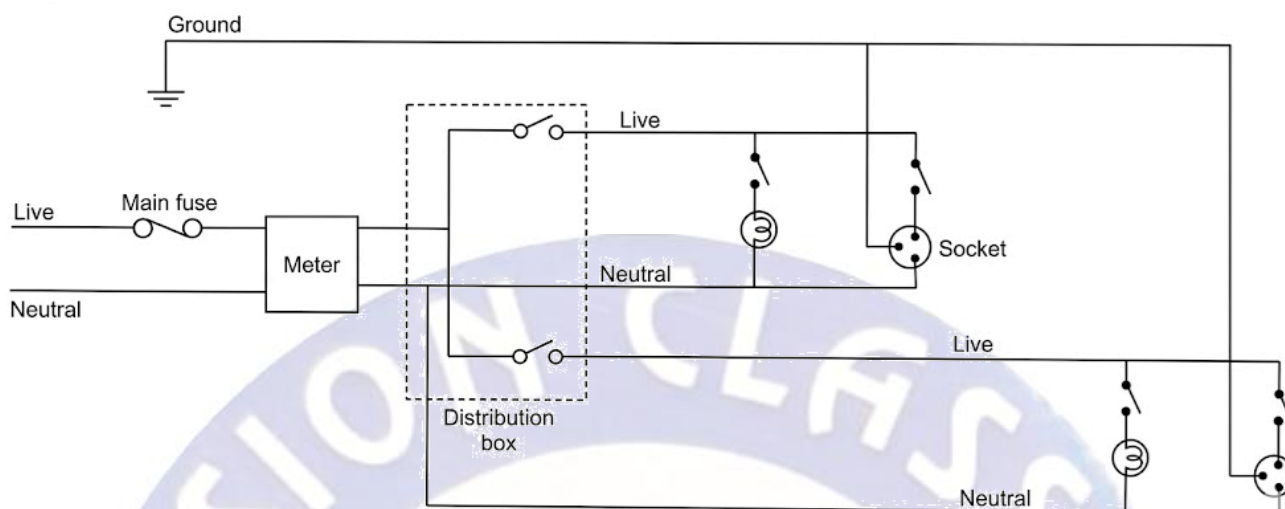
I = amount of electric current

R = radius of circle

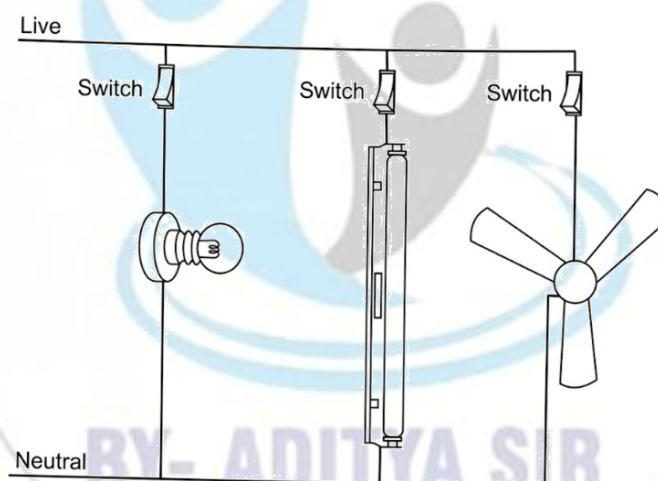


- **Domestic electric current OR Domestic wiring**

- **Domestic electric circuits are the circuits that are used in homes to supply electricity throughout our base. They are used to power a variety of houses hold appliances and these circuit light up our home.**

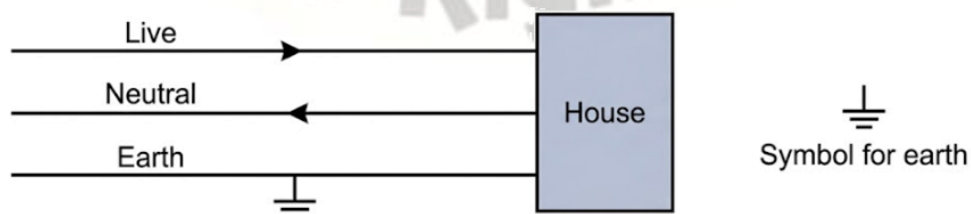


Wiring for two sections of a house



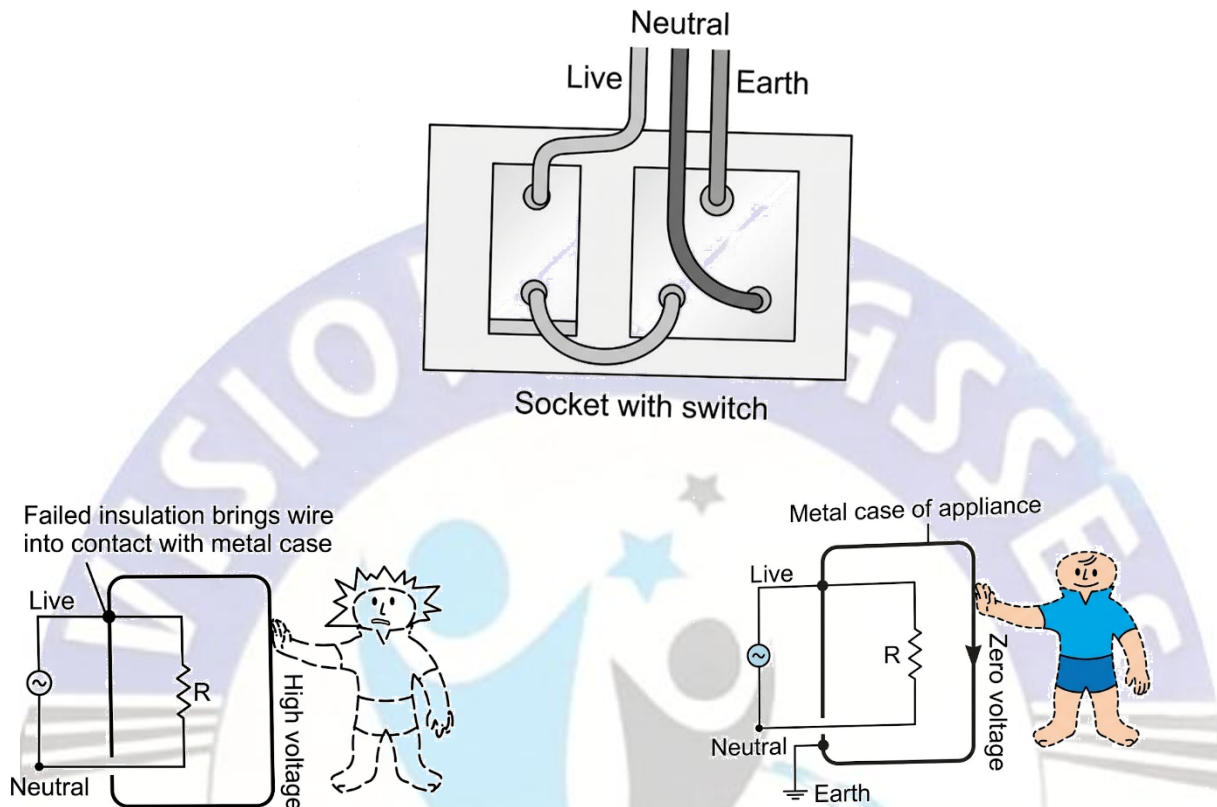
- **There are three types of wire is used in domestic electric current**

- Green wire [earth wire].**
- Red wire [main wire].**
- Black wire [neutral wire].**



- Q. What are the uses of Green wire.**

Ans Green wire stands for grounding and earthing in metallic covered device. These wires is mainly used for high voltage moving appliances. The earth wire is used to protect a person from electric shock in case of any faults in circuit.



Q. Red wire & Black wire.

Ans According to the electrical standards of India, Red is used for the live (or main) power conductor, black wire is covered the neutral wire.

Q. What is Overloading?

Ans Overloading occurs when a large number of high-power electrical appliances (like an electric iron, heater, and air conditioner) are switched on at the same time, drawing an excessive amount of current from a single circuit.

The magnetic effect of current increases with current, and high current causes the wire to overheat. Overheating can melt insulation and start a fire.

Q. What is Short Circuit?

Ans A Short Circuit occurs when the live wire (which carries current) and the neutral wire come into direct contact with each other.

Resistance nearly disappears. Suddenly, a very large current flows, producing a strong magnetic effect. Wires may spark, overheat, or catch fire. Reasons for short circuits Insulation that is worn out or damaged Insecure connections Water getting inside electrical appliances Inadequate wiring.

Q. Why do we use electric fuse in electric circuit?

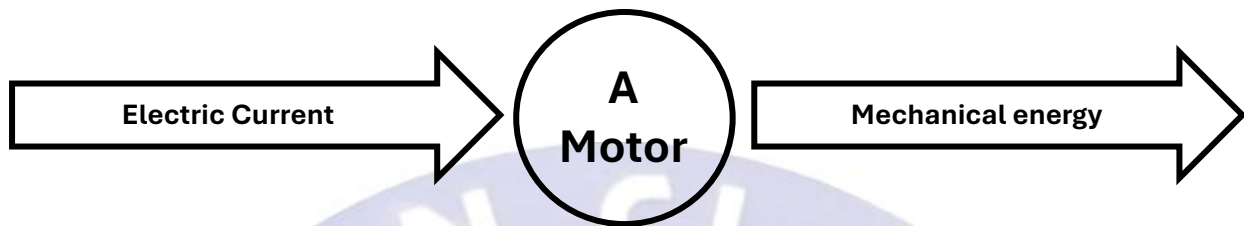
Ans An electric fuse protects the circuit and appliances from damage due to excess current.

When too much current flows (because of overloading or short circuit), the fuse melts and breaks the circuit.

This stops the flow of electricity, preventing overheating, fire, or damage to wires and devices.

▪ **In the case of electric motor**

- We know that the current carrying conductor experiences a force in a magnetic field and this experience force acts on the coil which rotates it continuously K.E.



▪ **In the case of generator or dynamo**

- We know that the changing magnetic field induces a current in a conductor and this induced current produced by generator.



• **DC motor / Electric motor**

Example:-

- ✓ The electric motor is used in fans, washing machine, refrigerator, mixer, grinder, electric cars etc. works on direct current, so it is called D.C motors OR direct current motors or motors.

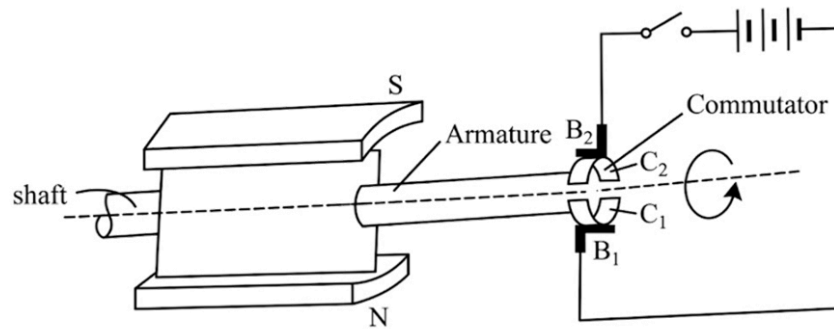
Construction:-

- A motor basically consists of a coil placed in a magnetic field. The coil is formed by winding many turns of insulated copper wire over a soft iron core. The core and the coil together form the armature. The armature is fixed on a rotating long cylinder rod, known as shaft of the motor and kept b/w [between] the poles of a strong magnet. The two free ends of the coil are connected to the two halves of a metallic split ring is called split ring commutator or commutator. When the shaft rotates, the commutator and the coil rotate with it.

Two carbon brushes B_1 and B_2 press against the two halves C_1 and C_2 of the commutator. A potential difference is applied across the brushes by connecting them to a source of DC, for example a battery.

Note:

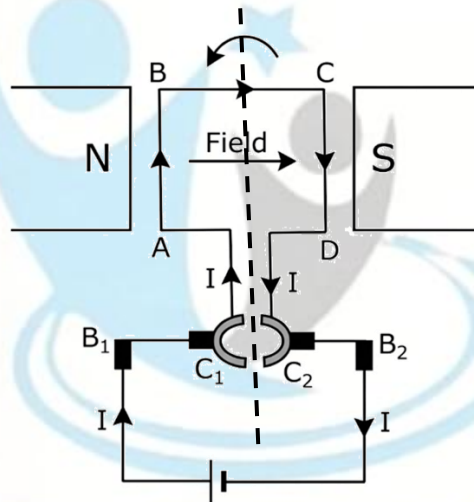
- We can't supply electric current (D.C) directly with connecting wires because the connecting wires will get twisted when the coil rotates and finally wire or broken.



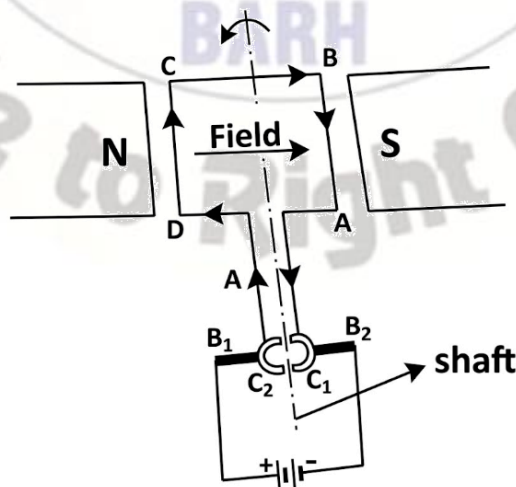
Electric motor

- **Working rule of electric motor**

- The brush B_1 is connected to the positive terminal of the battery and B_2 is connected to negative terminal. So, a current flows in the coil along ABCD ($A \rightarrow B \rightarrow C$).



after half rotation the commutator... get interchanged so current flows along DCBA



the commutator changes the direction of the current in the coil after each half-rotation. now Using Fleming's left-hand rule, we find that the arm AB now experiences an upward force, and the arm CD experiences a downward force. So, the coil continues to rotate in the anticlockwise direction.

Note:

- If the commutator is not used, the direction of current in an arm of the coil will always remain the same. So, the direction of the force on arm will not change and the coil will not be able to complete one rotation.

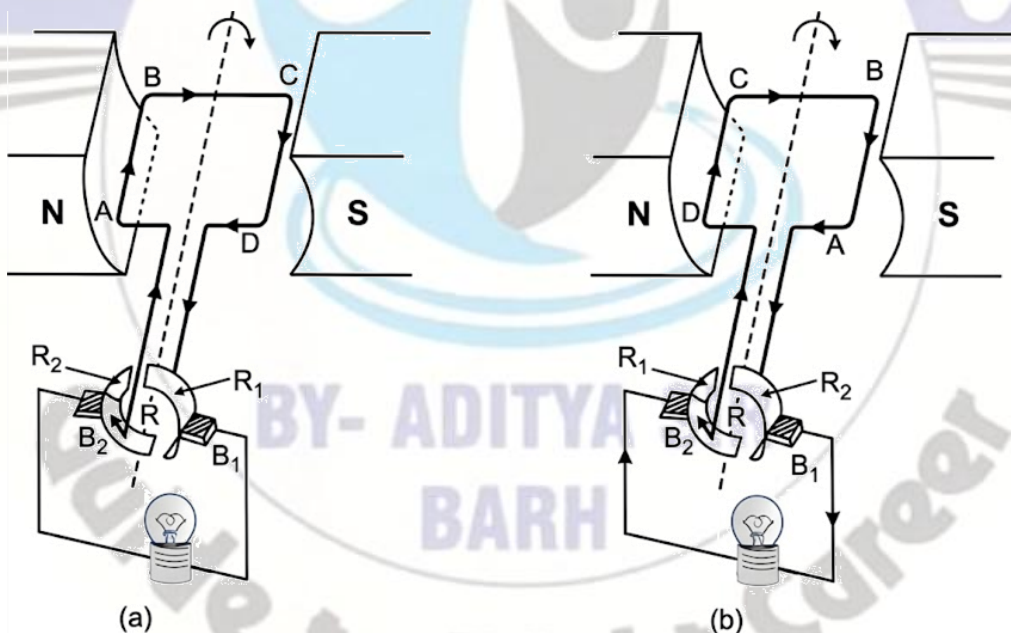
Dynamo / Generator

D.C Dynamo

A.C Dynamo

1. D.C Dynamo

- D.C Dynamo is similar to a D.C motor.
- The armature of the generator consists of a coil of many turns wound over a soft iron core. The armature is placed b/w [between] the poles of strong magnet. The ends of the coil are connected to a split-ring commutator. Two Carbon brushes, B₁ and B₂ press tightly against the two halves of the commutator. The brushes are connected to the fixed terminals, P and Q of the generator. Electric devices can be connected terminals.



▪ **Working of D.C generator / dynamo**

- To generate the electricity. The armature must be rotated in the magnetic field by the help of like as diesel engine. The circuit is connected with two ends of coil. A current flow in it. If the terminals are left open, there is no current but a potential difference appears b/w [between] them.

The brushes B₁ and B₂ are in contact with the halves C₁ and C₂ of the commutator respectively. B₁ & B₂ are connected to terminals P and Q. The arm AB of the coil is moving downwards, while the arms DC is moving upwards. The current passes through along DCBA. but after half a rotation, the brushes with which the two halves of the commutator were in contact get interchanged. Now, arm AB is moving upwards and arm DC is moving downwards This causes the current to flow along ABCD.

Note:

- D.C generator produces D.C current.

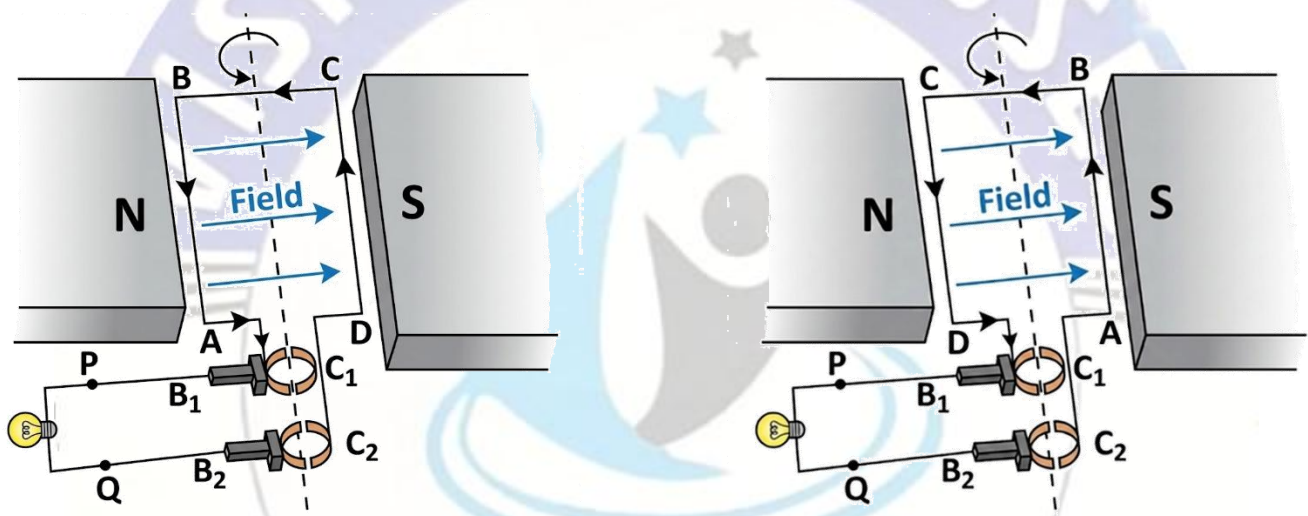
2. A.C Dynamo

→ means "Alternating Current Dynamo" because A.C Dynamo produce electric current.

➤ A.C generator is also called an alternator.

Construction:-

➤ A simple A.C generator consists of a rectangular coil ABCD of many turns wound over a soft iron core. The armature is placed between the poles of a strong magnet. The ends of the coil are connected to two cylindrical slip rings, C_1 and C_2 , mounted on the shaft. Each slip ring is permanently in contact with a carbon brush. The brushes are connected to the fixed terminals, p and q. Electric devices such as an electric bulb can be connected to these terminals as shown in the figure.



▪ Working rule of an A.C dynamo

➤ To generate electricity, the shaft on which the coil is mounted, rotated, the arm AB of the coil is moving down, while the arm is moving up.

A/c [According] to Fleming's Right-hand rule will show that the current in the coil is along DCBA. From A, the current flows to D.

The situation after half a rotation. the arm AB is now moving up, and the arm DC is moving down. This causes the current flow along ABCD. From D to A.

Note:

➤ The major difference b/w [between] A.C & D.C dynamo is Ring commutator in A.C Dynamo & half Commutator in D.C Dynamo.